

Examiner — Complete Build Process & Technology Stack

1. Project Introduction

Project Name

Examiner

Project Type

AI-Powered Virtual Interview Platform

Project Overview

Examiner is a Full Stack LLM-powered conversational AI interview platform designed to simulate realistic technical interviews.

The platform uses:

- Generative AI
- NLP
- Speech Processing
- Computer Vision
- Real-time AI interaction

to create a professional interview experience for candidates.

Main Goal

The main objective of Examiner is to help students prepare for:

- technical interviews
- HR interviews
- coding rounds
- communication rounds

through AI-powered real-time interview simulation.

2. Core Features of Examiner

Feature 1 — AI Interview Panel

Instead of a single interviewer:

Examiner contains:

Three Animated AI Interviewers

The system simulates a realistic interview panel.

Example Interview Panel

Interviewer 1

Technical Interviewer

Purpose:

- technical questions
 - coding concepts
 - subject evaluation
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Interviewer 2

HR Interviewer

Purpose:

- communication analysis
 - behavioral questions
 - confidence evaluation
-

Interviewer 3

Domain Expert

Purpose:

- advanced follow-up questions
- deep technical analysis

- project discussion
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Why Three Interviewers?

Advantages:

- realistic interview environment
- better interaction
- panel interview simulation
- improved immersion
- dynamic conversation experience

This becomes a major innovation point of Examiner.

Feature 2 — Two Interview Modes

Candidate gets:

OPTION 1 — Resume-Based Interview

Candidate uploads resume.

System:

- extracts resume content
- analyzes skills/projects
- generates personalized interview questions

Example:

Resume Contains:

- Java
- Spring Boot
- React
- AI Project

AI generates:

Explain your Spring Boot project architecture.

OPTION 2 — Subject-Based Interview

Candidate enters subject.

Example:

Java

System generates:

- beginner questions
- medium questions
- advanced questions

based on selected difficulty.

3. Problem Statement

Existing Problems

Current interview preparation systems:

- provide static MCQs
- provide fixed questions
- lack human interaction
- cannot evaluate communication
- cannot analyze confidence
- cannot simulate real interview environment

Because of this:

- students become nervous
 - communication skills remain weak
 - interview confidence decreases
 - practical interview exposure is missing
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4. Proposed Solution

Examiner solves these problems by providing:

✓ AI-generated interview questions ✓ Dynamic conversation ✓ Real-time evaluation ✓ Resume-based interview generation ✓ Subject-based interview generation ✓ Confidence analysis ✓ Communication analysis ✓ AI interview panel simulation ✓ Final performance analytics

5. Complete System Workflow

Step 1 — Candidate Opens Platform

Landing page displays:

- three animated interviewers
 - futuristic interview environment
 - subject input option
 - resume upload option
-

Step 2 — Candidate Selects Interview Mode

Mode A — Resume Interview

Candidate uploads resume.

System extracts:

- skills
 - projects
 - technologies
 - experience
-

Mode B — Subject Interview

Candidate enters:

Java

Candidate selects:

- beginner
 - medium
 - advanced
-

Step 3 — Interview Initialization

Spring Boot backend creates:

- interview session
 - candidate context
 - interview state
 - AI prompt context
-

Step 4 — AI Interview Begins

Gemini API generates:

- technical questions
- HR questions
- follow-up questions

Each animated interviewer performs specific roles.

Step 5 — Candidate Response

Candidate responds using:

- text OR
 - voice
-

Step 6 — AI Processing

System processes:

NLP Analysis

Analyzes:

- communication
 - sentence quality
 - technical correctness
-

Speech Analysis

Analyzes:

- confidence
 - hesitation
 - voice stability
-

Computer Vision Analysis

Analyzes:

- eye contact
 - posture
 - facial emotion
 - attention level
-

Step 7 — Dynamic AI Follow-Up

Based on performance:

AI:

- changes difficulty
 - asks follow-up questions
 - evaluates depth of knowledge
-

Step 8 — Final Report Generation

System generates:

✓ Technical Score ✓ Communication Score ✓ Confidence Score ✓ Weak Areas ✓ Strong Areas ✓
Suggested Improvements

6. System Architecture

Complete Architecture

Frontend (React)

↓

Spring Boot Backend

```
↓
Gemini API
↓
Python ML Service
↓
NLP + CV + Speech Models
```

7. Required Project Folders

```
Examiner/
|
├── Examiner-frontend/
├── Examiner-backend/
└── Examiner-ml/
```

8. Responsibilities of Each Folder

A. Examiner-frontend

Purpose

User Interface Layer

Responsibilities

Frontend handles:

- landing page
 - animated AI interviewers
 - interview UI
 - resume upload
 - subject input
 - voice interaction
 - webcam access
 - final report dashboard
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Technologies Used

- React
 - Tailwind CSS
 - Framer Motion
 - React Three Fiber
 - Three.js
-

B. Examiner-backend

Purpose

Main System Controller

Responsibilities

Backend handles:

- authentication
 - interview sessions
 - Gemini API communication
 - report generation
 - resume processing
 - frontend communication
 - orchestration
-

Technologies Used

- Spring Boot
 - Spring Security
 - JWT
 - WebSocket
 - MongoDB/PostgreSQL
-

C. Examiner-ml

Purpose

AI Processing Layer

Responsibilities

ML service handles:

- speech-to-text
- emotion detection
- confidence analysis
- eye tracking
- NLP analysis
- communication analysis

Technologies Used

- Python
- FastAPI
- OpenCV
- MediaPipe
- Whisper
- DeepFace
- Librosa
- spaCy

9. Technology Stack

Frontend Stack

Technology	Purpose
React	Frontend framework
Tailwind CSS	UI styling
Framer Motion	UI animation
React Three Fiber	3D rendering
Three.js	3D interviewer models

Backend Stack

Technology	Purpose
Spring Boot	Backend framework

Technology	Purpose
Spring Security	Authentication
JWT	Secure login
WebSocket	Real-time communication
MongoDB/PostgreSQL	Database

AI & ML Stack

Tool/Model	Purpose
Gemini API	AI interview intelligence
Whisper	Speech-to-text
Coqui TTS	AI voice generation
OpenCV	Webcam analysis
MediaPipe	Eye/head tracking
DeepFace	Emotion detection
Librosa	Confidence analysis
spaCy	NLP analysis

10. Communication Between Components

FRONTEND → BACKEND

Frontend sends:

- candidate input
- voice input
- webcam frames
- resume files

BACKEND → GEMINI API

Backend sends:

- prompts
- candidate responses

- interview history

Gemini returns:

- interview questions
 - evaluations
 - follow-up questions
-

BACKEND → ML SERVICE

Backend sends:

- audio data
- webcam data
- candidate communication

ML Service analyzes:

- confidence
 - emotion
 - communication
 - posture
-

ML SERVICE → BACKEND

ML service returns:

- confidence score
 - emotion analysis
 - communication metrics
 - attention tracking
-

BACKEND → FRONTEND

Backend sends:

- AI responses
 - interview questions
 - live updates
 - final analytics report
-

11. Build Process of Examiner

PHASE 1 — MVP Development

Build:

- landing page
 - three animated interviewers
 - subject input
 - resume upload
 - Gemini integration
 - text interview system
 - report generation
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PHASE 2 — Voice AI Integration

Add:

- Whisper
 - Coqui TTS
 - real-time AI speaking
 - voice interview interaction
-

PHASE 3 — AI Behavioral Analysis

Add:

- OpenCV
- MediaPipe
- DeepFace
- Librosa

Features:

- confidence detection
 - eye tracking
 - emotion analysis
 - posture analysis
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PHASE 4 — Advanced Features

Add:

- coding interview mode
 - company-specific interviews
 - AI analytics dashboard
 - interview history
 - performance tracking
-

12. Innovation Points of Examiner

Main Innovations

✓ Three AI interviewer panel ✓ Resume-based interview generation ✓ Subject-based interview generation ✓ Dynamic AI questioning ✓ Real-time conversational interview ✓ Confidence analysis ✓ Emotion detection ✓ Communication evaluation ✓ Realistic interview simulation

13. Professional Category of Project

Examiner belongs to:

- Full Stack AI Development
 - Conversational AI Systems
 - Generative AI Applications
 - LLM-Powered Platforms
 - Multimodal AI Systems
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14. Final Conclusion

Examiner is a Full Stack LLM-powered conversational AI interview platform that combines Generative AI, NLP, Speech Processing, and Computer Vision technologies to create realistic AI-based interview simulations.

The system provides:

- personalized interviews
- resume-based questioning
- AI panel interaction
- confidence analysis
- communication evaluation
- real-time conversational experience

through a scalable Spring Boot + AI architecture.